

3

Regression, Observations, and Interventions

In this chapter, we're going to build a link between **associations**, **interventions**, and **regression** models. We'll look into the logic of **statistical control** – a tool used by scientists in the hopes of making their models more robust. Finally, we'll look into the connection between regression and structural models.

By the end of this chapter, you should have a solid understanding of statistical control and how it can help in estimating causal effects from observational data. This knowledge will allow us to build the more complex non-linear models introduced in *Part 2, Causal Inference*.

In this chapter, we'll cover the following topics:

- Associations in observational data versus linear regression
- Causal perspective on statistical control
- Regression and structural models

Starting simple – observational data and linear regression

In previous chapters, we discussed the concept of association. In this section, we'll quantify associations between variables using a regression model. We'll see the geometrical interpretation of this model and demonstrate that regression can be performed in an arbitrary direction. For the sake of simplicity, we'll focus our attention on linear cases. Let's start!

Linear regression

Linear regression is a basic data-fitting algorithm that can be used to predict the expected value of a dependent (target) variable, Y , given values of some predictor(s), X . Formally, this is written as $\hat{Y}_{X=x} = E[Y|X = x]$.

In the preceding formula, $\hat{Y}_{X=x}$ is the predicted value of Y given that X takes the value(s) x . $E[\cdot]$ is the **expected value** operator. Note that X can be multidimensional. In such cases, X is usually represented as a matrix, $\mathbf{X}$, with shape $N \times D$, where N is the number of observations and D is the dimensionality of X (the number of predictor variables). We call the regression model with multidimensional X a **multiple regression**.

An important feature of linear regression is that it allows us to easily quantify the strength of the relationship between predictors and the target variable by computing regression **coefficients**. Intuitively, regression coefficients can be thought of as the amount of change in the *predicted* output variable relative to a unit change in the input variable.

Coefficients and multiple regression

In multiple regression with k predictors $X_1, \dots, X_k$, each predictor, X_j , has its respective coefficient, β_j . Each coefficient, β_j , represents the *relative* contribution of X_j to the change in the predicted target, $\hat{Y}$, holding everything else *constant*.

Let's take a model with just one predictor, X . Such a model can be described by the following formula:

$$\hat{y}_i = \alpha + \beta x_i$$

In the preceding formula, $\hat{y}_i$ is a predicted value for observation i , α is a learned **intercept** term, x_i is the observed value of X , and β is the regression coefficient for X . We call α and β model parameters.

Let's build a simple example:

1. First, we'll define our data-generating process. We'll make the process follow the (preceding) linear regression formula and assign arbitrary values to the (true) parameters α^* and β^* . We'll choose 1.12 as the value of α^* and 0.93 as the value of β^* (you can use other values if you want). We will also add noise to the model and mark it as ϵ . We choose ϵ to be normally distributed with zero mean and a standard deviation of one. Additionally, we'll scale ϵ by 0.5 . With these values, our data-generating formula becomes the following:

$$y_i = 1.12 + 0.93 x_i + 0.5 \epsilon_i$$

Non-linear associations and interactions

Non-linear associations are also quantifiable. Even linear regression can be used to model some non-linear relationships. This is possible because linear regression has to be *linear in parameters*, not necessarily in the data. More complex relationships can be quantified using entropy-based metrics such as mutual information (Murphy, 2022; pp. 213-218). Linear models can also handle **interaction** terms. We talk about interaction when the model's output depends on a multiplicative relationship between two or more variables. In linear regression, we model the interaction between two predictors by adding an additional multiplicative term to the equation. For instance, if we want to model an interaction between two features, X_1 and X_2 , we add the multiplication of X_1 and X_2 to the equation: $Y = X_1 + X_2 + X_1 X_2$.

2. Next, we'll translate our data-generating formula into code in order to generate the data. The code for this chapter is in the `Chapter_03.ipynb` notebook (<https://bit.ly/causal-ntbk-03>).

Let's put it to work!

1. We'll start by importing the libraries that we're going to use in this chapter. We're going to use `statsmodels` to fit our linear regression model:

```
import numpy as np
import statsmodels.api as sm
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
```

statsmodels

`statsmodels` is a popular statistical library in Python that offers support for R-like syntax and R-like model summaries (in case you haven't heard of R, it is a popular open source statistical programming language). `statsmodels` is a great choice if you want to work with traditional statistical models. The package offers convenient model summaries that contain p -values and other useful statistics. If you come from a Scikit-learn background, you might find the `statsmodels` API a bit confusing. There are several key differences between the two libraries. One of them is the `.fit()` method, which in `statsmodels` returns an instance of a wrapper object that can be further used to generate predictions. For more details on `statsmodels`, refer to the documentation: <https://bit.ly/StatsmodelsDocs>.

2. Next, we'll set a random seed for reproducibility and define the number of samples that we're going to generate:

```
np.random.seed(45)
N_SAMPLES = 5000
```

3. Then, we'll define our model parameters, α^* and β^* :

```
alpha = 1.12
beta = 0.93
epsilon = np.random.randn(N_SAMPLES)
```

4. Finally, we'll use our model formula to generate the data:

```
X = np.random.randn(N_SAMPLES)
y = alpha + beta * X + 0.5 * epsilon
```

5. There's one more step that we need to take before fitting the model. `statsmodels` requires us to add a constant feature to the data. This is needed to perform the intercept computations. Many libraries perform this step implicitly; nonetheless, `statsmodels` wants us to do it

explicitly. To make our lives easier, the authors have provided us with a convenient method, `.add_constant()`. Let's apply it!

```
X = sm.add_constant(X)
```

Now, our `X` has got an extra column of ones at column index 0. Let's print the first five rows of `X` to see it:

```
print(X[:5, :])
```

The result is as follows:

```
[ 1.      0.11530002]
[ 1.     -0.43617719]
[ 1.     -0.54138887]
[ 1.     -1.64773122]
[ 1.     -0.32616934]
```

Now, we're ready to fit the regression model using `statsmodels` and print the summary:

```
model = sm.OLS(y, X)
fitted_model = model.fit()
print(fitted_model.summary())
```

The output of the model summary is presented in *Figure 3.1*. We marked the estimated coefficients with a red ellipse:

OLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.771			
Model:	OLS	Adj. R-squared:	0.771			
Method:	Least Squares	F-statistic:	1.681e+04			
Date:	Sun, 05 Jun 2022	Prob (F-statistic):	0.00			
Time:	10:51:22	Log-Likelihood:	-3615.0			
No. Observations:	5000	AIC:	7234.			
Df Residuals:	4998	BIC:	7247.			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	1.1243	0.007	159.391	0.000	1.110	1.138
x1	0.9212	0.007	129.669	0.000	0.907	0.935
=====						
Omnibus:	0.129	Durbin-Watson:	1.986			
Prob(Omnibus):	0.938	Jarque-Bera (JB):	0.163			
Skew:	0.002	Prob(JB):	0.922			
Kurtosis:	2.972	Cond. No.	1.02			
=====						

Figure 3.1 – A summary of the results of a simple linear regression model

The coefficient marked `const` is α , the estimate of the true α^* parameter, while the coefficient marked `x1` is β , the estimate of β^* . They are slightly different from their true counterparts ($\alpha^* = 1.12$, $\beta^* = 0.93$). This is because we made our model noisy by adding the ϵ term. You can see that both coefficients are associated with p -values below 0.001 (check the column named $P > |t|$), which indicates that they are **statistically significant** at the customary $p < .05$ level.

p-values and statistical significance

Broadly speaking, the p -value is a statistical device meant to help distinguish between the signal and the noise in statistical comparisons or summaries. More precisely, the p -value is the probability of observing data at least as extreme as we observed, given that the *null hypothesis* is true.

The **null hypothesis** usually states that there is *no effect* or *no difference* between two or more objects that we compare. In the context of linear regression, we test two types of null hypotheses:

- Null hypotheses for coefficients (including the intercept)
- A null hypothesis for the entire model

The null hypothesis for a given coefficient states that this coefficient is not significantly different from zero. The null hypothesis for the model states that the entire model is not significantly different from the **null model** (in the context of simple regression analysis, the null model is usually represented as an intercept-only model).

If you want to learn more about null hypotheses and hypothesis testing, check out this video series from Khan Academy: <https://bit.ly/KhanHypothesisTesting>.

Given a null hypothesis, p -values are used as a convenient quantitative summary that helps us understand whether we can safely reject this null hypothesis for some threshold value (a good practice is to pick the threshold value before starting the analysis or – preferably – before starting the data collection process).

Because p -values are the probability of observing data *at least as extreme* as ours under the null hypothesis, the lower the p -value, the *less likely* the null hypothesis is. In other words, the lower the p -value, the *less comfortable* we are in holding that the null hypothesis is true, and the *more likely* we are to agree that it should be *rejected*. If the p -value is lower than the threshold that we picked, we say that we *reject the null hypothesis* (note that we don't say that we've proven an alternative to the null hypothesis).

Although p -values have been widely adopted in modern statistics and science, they have also been widely abused. This has led to multiple severe critiques of p -values and statistical significance, highlighting that they are frequently misused and misinterpreted, which can lead to detrimental real-world consequences. For a critical overview, check out Wasserstein and Lazar (2016).

Geometric interpretation of linear regression

Linear regression can also be viewed from a geometric point of view. Let's plot the data we generated alongside the fitted regression line:

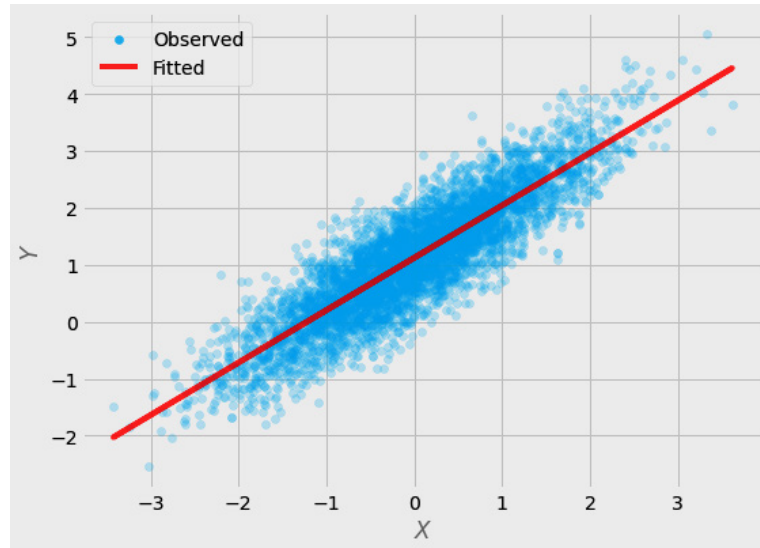


Figure 3.2 – Generated data and fitted regression line

Each blue dot in *Figure 3.2* represents a single observation, while the red line represents the best-fit line found by the linear regression algorithm. In the case of multiple regression, the line becomes a *hyperplane*.

Regression and correlation

If we imagine the plot in *Figure 3.2* without the red line, we are still able to recognize a strong linear relationship between X and Y . This might remind some of you of the concept of correlation. In fact, there are a number of similarities between Pearson's correlation coefficient and linear regression. For instance, if we standardize a coefficient of X in the regression model $Y \sim X$, it will have the same value as Pearson's r between X and Y . There are also some differences. For instance, Pearson's r between X and Y will be the same as Pearson's r between Y and X , yet unstandardized coefficients in regression models $Y \sim X$ and $X \sim Y$ will typically differ.

Reversing the order

Regression is a purely statistical *rung 1* model and we can use it to quantify the association between X and Y as well as between Y and X . In other words, regression does not say anything about the data's causal structure. In particular, there might be no causal link between two variables at all but we can still find a relationship between them using a regression model.

In the first chapter, we discussed the example of an association between ice cream sales and drownings. We showed that this association was *spurious*, but the regression model would still quantify it as existing (assuming that the association would be strong enough and possible to express using the model of choice, in our case *linear* regression). This is the nature of the first rung of *The Ladder of Causation* and – as we mentioned in *Chapter 2* – it can be very useful in certain cases.

More on regression vocabulary

When we use X as a predictor and Y as a target variable (also called the **dependent variable**), we say that we regress Y on X . If we use Y as a predictor of X , we say that we regress X on Y . Regressing Y on X can be also expressed in R-style notation as $Y \sim X$, and regressing X on Y as $X \sim Y$. We will use this notation across the book to describe various models. We decided to use R-style notation because of its neatness and simplicity. In some cases, though, it might be clearer to just say that we *regress Y on X* rather than using a formula. We'll use this descriptive style where clarity demands it.

To make it more hands-on, let's see what the reversed regression model looks like. We will now regress X on Y . The code to build the reversed model is very similar to the code to build the original model, so we won't discuss it here. If you want to examine the code for yourself, you can find it in the `Chapter_03.ipynb` notebook (<https://bit.ly/causal-ntbk-03>).

Let's take a look at the results summary in *Figure 3.3*:

OLS Regression Results						
Dep. Variable:	y	R-squared:	0.771			
Model:	OLS	Adj. R-squared:	0.771			
Method:	Least Squares	F-statistic:	1.681e+04			
Date:	Sun, 05 Jun 2022	Prob (F-statistic):	0.00			
Time:	19:16:00	Log-Likelihood:	-3375.0			
No. Observations:	5000	AIC:	6754.			
Df Residuals:	4998	BIC:	6767.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-0.9441	0.010	-96.048	0.000	-0.963	-0.925
x1	0.8368	0.006	129.669	0.000	0.824	0.849
Omnibus:	0.590	Durbin-Watson:	1.994			
Prob(Omnibus):	0.745	Jarque-Bera (JB):	0.582			
Skew:	-0.026	Prob(JB):	0.748			
Kurtosis:	3.003	Cond. No.	2.83			

Figure 3.3 – Results summary for the reversed model

As we can see, the coefficients have changed. In particular, the *intercept* is now negative. As we're now regressing X on Y , the intercept became the point where the fitted line crosses the X axis (rather than the Y axis as in the original model). You can verify that the fitted line crosses the X axis below 0 by looking at Figure 3.4 (the reversed model) and Figure 3.2 (the original model):

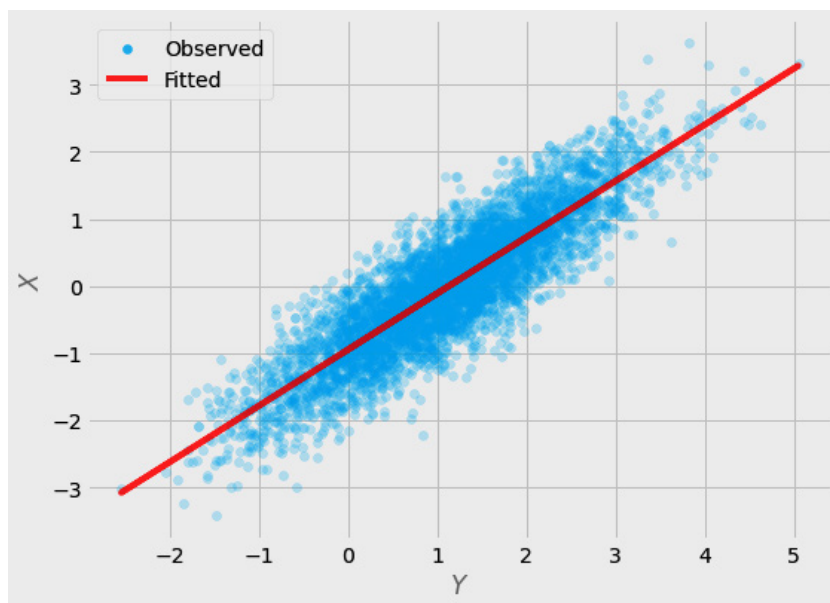


Figure 3.4 – Visualization of the reversed model

The regression model itself cannot help us understand which variable is the cause and which is the effect. To determine this, we need some sort of external knowledge.

Causal attributions become even more complicated in *multiple regression*, where each additional predictor can influence the relationship between the variables in the model. For instance, the learned coefficient for variable X might be 0.63, but when we add variable Z to the model, the coefficient for X changes to -2.34. A natural question in such cases is: if the coefficient has changed, what is the true effect here?

Let's take a look at this issue from the point of view of statistical control.

Should we always control for all available covariates?

Multiple regression provides scientists and analysts with a tool to perform **statistical control** – a procedure to remove *unwanted* influence from certain variables in the model. In this section, we'll discuss different perspectives on statistical control and build an intuition as to why statistical control can easily lead us astray.

Let's start with an example. When studying predictors of dyslexia, you might be interested in understanding whether parents smoking influences the risk of dyslexia in their children. In your model, you might want to control for parental education. Parental education might affect how much attention parents devote to their children's reading and writing, and this in turn can impact children's skills and other characteristics. At the same time, education level might decrease the probability of smoking, potentially leading to confounding. But how do we actually know whether it *does* lead to confounding?

In some cases, we can refer to previous research to find an answer or at least a hint. In other cases, we can rely on our intuition or knowledge about the world (for example, we know that a child's current skills cannot cause the parent's past education). However, in many cases, we will be left without a clear answer. This inevitable uncertainty led to the development of various heuristics guiding the choice of variables that should be included as statistical controls.

Navigating the maze

One of the existing heuristics is to control for as many variables as possible. This idea is based on *"the (...) assumption that adding CVs [control variables] necessarily produces more conservative tests of hypotheses"* (in: Becker et al., 2016). Unfortunately, this is not true. Moreover, controlling for wrong variables can lead to severely distorted results, including spurious effects and reversed effect signs.

Some authors offer more fine-grained heuristics. For example, Becker and colleagues (Becker et al., 2016; <https://bit.ly/BeckersPaper>) shared a set of 10 recommendations on how to approach statistical control. Some of their recommendations are as follows (the original ordering is given in parentheses):

- If you are not sure about a variable, don't use it as a control (1)
- Use conceptually meaningful control variables (3)
- Conduct comparative tests of relationships between the independent variables and control variables (7)
- Run results with and without the control variables and contrast the findings (8)

We'll see a couple of examples of scenarios leading to confounding in a while, but first, let's take a look at some of the author's recommendations.

If you don't know where you're going, you might end up somewhere else

Recommendations (1) and (3) discourage adding variables to the model. This might sound reasonable – if you're not sure, don't add, because you might break something by accident. It seems rational, perhaps because most of us have seen or participated in situations like this in real life – someone does not understand how something works, they do something that seems sensible to them, but they are not aware of their own blind spots and the thing is now broken.

An important aspect of this story is that the thing is *now* broken. This suggests that it *worked properly before*. This is not necessarily a valid assumption to make from a causal point of view. *Not including* a variable in the model might also lead to *confounding* and *spuriousness*. This is because there are various patterns of **independence structure** possible between any three variables. Let's consider the **structural causal model (SCM)** presented in Figure 3.5:

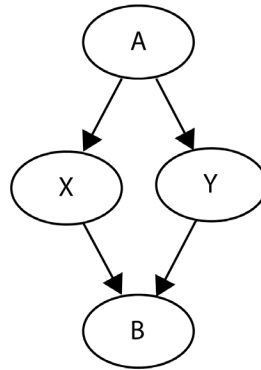


Figure 3.5 – An example SCM with various patterns leading to spurious associations

More on SCM graphical representations

The representation of the SCM in Figure 3.5 differs slightly from the representations we used in the previous chapter. First, it does not contain noise variables. You can think of this representation as a simplified version with *implicit* noise variables – clearer and more focused. Second, nodes are represented as ellipses rather than circles. This is because we used the default mode of the `graphviz` library to generate them. The difference in the shapes does not have any particular meaning *in our case*. However, this is a great opportunity to introduce `graphviz` – a software package and a Python library for graph visualization. It's a useful tool when you work with causal models and smaller graphs or networks. To learn the basic `graphviz` syntax, you can refer to the notebooks accompanying this book. For a more comprehensive introduction, check out <https://bit.ly/GraphvizDocs>.

From the model structure, we can clearly see that X and Y are *causally* independent. There's no arrow between them, nor is there a directed path that would connect them indirectly. Let's fit four models and analyze which variables, when controlled for, lead to spurious relationships between X and Y :

1. First, we'll start simple with a model that regresses Y on X .
2. Then, we'll add A to this model.
3. Next, we'll fit a model without A , but with B .
4. Finally, we'll build a model with all four variables.

What is your best guess – out of the four models, which ones will correctly capture causal independence between X and Y ? I encourage you to write your hypotheses down on a piece of paper before we reveal the answer. Ready? Let's find out!

The code for this experiment is in the `Chapter_03.ipynb` notebook (<https://bit.ly/causal-ntbk-03>). Follow these steps:

1. First, let's define the SCM:

```
a = np.random.randn(N_SAMPLES)
x = 2 * a + 0.5 * np.random.randn(N_SAMPLES)
y = 2 * a + 0.5 * np.random.randn(N_SAMPLES)
b = 1.5 * x + 0.75 * y
```

Note that all the coefficients that we use to scale the variables are arbitrarily chosen.

2. Next, let's define four model variants and fit the models iteratively:

```
# Define four model variants
variants = [
    [x],
    [x, a],
    [x, b],
    [x, a, b]
]

# Fit models iteratively and store the results
for variant in variants:
    X = sm.add_constant(np.stack(variant).T)
    model = sm.OLS(y, X)
    fitted_model = model.fit()
```

3. Finally, let's examine the results in *Table 3.1*:

Model	X		A		B	
	p-value	Coefficient	p-value	Coefficient	p-value	Coefficient
$Y \sim X$	< .001	0.947	-	-	-	-
$Y \sim X + A$	.6565	0.014	<.001	1.967	-	-
$Y \sim X + B$	< .001	-2.000	-	-	< .001	1.333
$Y \sim X + A + B$	< .001	-2.000	<.001	0.000	< .001	1.333

Table 3.1 – Summary of the results of four regression models

What we see in *Table 3.1* is that the only model that recognized the causal independence of X and Y correctly (large p -value for X , suggesting the lack of significance) is the second model ($Y \sim X + A$). This clearly shows us that all other statistical control schemes led to invalid results, including the model that does not control for *any* additional variables.

Why did controlling for A work while all other schemes did not? There are three elements to the answer:

- First, A is a confounder between X and Y and we need to control for it in order to remove confounding. This situation is structurally identical to the one in the ice cream example in the first chapter.
- Second, X , Y , and B form a pattern that we call a **collider** or **immorality**. This pattern has a very interesting behavior – it enables the flow of information between the parent variables (X and Y in our case) when we control for the child variable (B in our example). This is exactly the opposite of what happened when we controlled for A !
- Third, not controlling for *any* variable leads to the same result in terms of the *significance* of X as controlling for A and B (note that the results are different in terms of *coefficients*, yet as we're now interested in the structural properties of the system, this is of secondary importance). This is precisely because the effects of controlling for A and controlling for B are exactly the opposite from a *structural point of view* and they cancel each other out!

We'll devote the whole of *Chapter 5*, to a detailed discussion on the *collider* and two other graphical patterns (you already know their names, don't you?).

Get involved!

Now, let's get back to the recommendations given by Becker and colleagues. Recommendations (7) and (8) are interesting. Running comparative tests between variables can be immensely helpful in discovering causal relationships between them. Although Becker proposes to run these tests only between independent and control variables, we do not have to (and should not) restrict ourselves to this. In fact, comparative independence tests are an essential ingredient of some of the causal discovery methods that we'll discuss in *Part 3, Causal Discovery*.

To control or not to control?

The fact that smart people all over the world create heuristics to decide whether a given variable should be included in the model highlights how difficult it is to understand causal relationships between variables in complex and noisy real-world scenarios.

If we have full knowledge about the causal graph, the task of deciding which variables we should control for becomes relatively easy (and after reading the next couple of chapters, you might even find it almost trivial). If the true causal structure is unknown, the decision is fundamentally difficult.

There's no one-size-fits-all solution to the control-or-not-to-control question, but understanding causality will help you make much better decisions in this regard.

Causality does not give you a new angle on statistical control; it gives you new eyes that allow you to see what's *invisible* from the perspective of *rung 1* of *The Ladder of Causation*. For a summary of what constitutes good and bad controls, check out the excellent paper by Cinelli et al. (2022; <https://bit.ly/GoodBadControls>).

Regression and structural models

Before we conclude this chapter, let's take a look at the connection between regression and SCMs. You might already have an intuitive understanding that they are somehow related. In this section, we'll discuss the nature of this relationship.

SCMs

In the previous chapter, we learned that SCMs are a useful tool for encoding causal models. They consist of a set of variables (exogenous and endogenous) and a set of functions defining the relationships between these variables. We saw that SCMs can be represented as graphs, with nodes representing variables and directed edges representing functions. Finally, we learned that SCMs can produce interventional and counterfactual distributions.

SCM and structural equations

In causal literature, the names **structural equation model (SEM)** and **structural causal model (SCM)** are sometimes used interchangeably (e.g., Peters et al., 2017). Others refer to SEMs as a family of specific multivariate modeling techniques with latent variables (e.g., Bollen and Noble, 2011). SEM as a modeling technique is a vast and rich topic. For a good introduction, check out the book by Kline (2015); for Judea Pearl's account on SEM and causality, check out Pearl (2012).

Linear regression versus SCMs

Linear regression is a model that allows us to quantify the (relative) strength of a (linear in parameters) relationship between two or more variables. There is no notion of causal directionality in linear regression, and in this sense, we don't know which direction (if any) is the *causally correct* one. This condition is known as *observational equivalence* (Peters et al., 2017).

Finding the link

In the previous section, we used linear regression to estimate coefficients that we interpreted as *causal* estimates of the strength of a relationship between variables.

When fitting the four models describing the SCM from *Figure 3.5*, we saw that in the correct model ($Y \sim X + A$), the estimate of the coefficient for A was equal to 1.967. That's very close to the true coefficient, which was equal to 2. This result shows the direction of our conclusion.

Linear regression can be used to estimate causal effects, given that we know the underlying causal structure (which allows us to choose which variables we should control for) and that the underlying system is linear in terms of parameters.

Linear models can be a *useful microscope* for causal analysis (Pearl, 2013).

To cement our intuition regarding the link between linear regression and SCMs, let's build one more SCM that will be linear in terms of parameters but non-linear in terms of data and estimate its coefficients with linear regression:

1. As usual, let's first start by defining the causal structure. *Figure 3.6* presents a graphical representation of our model.

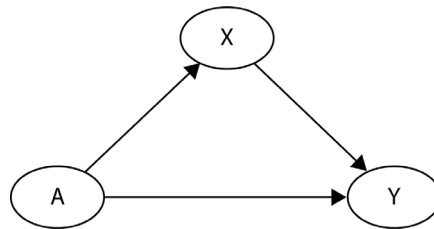


Figure 3.6 – Graphical representation of an example SCM

2. Next, let's define the functional assignments (we'll use the same settings for sample size and random seed as previously):

```
a = np.random.randn(N_SAMPLES)
x = 2 * a + .7 * np.random.randn(N_SAMPLES)
y = 2 * a + 3 * x + .75 * x**2
```

3. Let's add a constant and then initialize and fit the model:

```
X = sm.add_constant(np.stack([x, x**2, a]).T)
model = sm.OLS(y, X)
fitted_model = model.fit()
print(fitted_model.summary(xname=['const', 'x',
                                'x^2', 'a']))
```

Note that our functional assignment contained not only X but also X^2 . We want to make sure that we add the square term to the model as well. This is the simplest way to introduce non-linearity into a linear regression model. Also, note that the model is still linear in parameters (we only use addition and multiplication).

Another important thing to notice is that we included A in the model. The reason for this is that A is a confounder in our dataset and – as we learned before – we need to control for a confounder in order to get unbiased estimates.

Great, let's see the results!

OLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	1.000			
Model:	OLS	Adj. R-squared:	1.000			
Method:	Least Squares	F-statistic:	3.294e+32			
Date:	Sun, 05 Mar 2023	Prob (F-statistic):	0.00			
Time:	12:51:10	Log-Likelihood:	30899.			
No. Observations:	1000	AIC:	-6.179e+04			
Df Residuals:	996	BIC:	-6.177e+04			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-2.22e-16	3.57e-16	-0.622	0.534	-9.23e-16	4.79e-16
x	3.0000	4.14e-16	7.25e+15	0.000	3.000	3.000
x^2	0.7500	4.78e-17	1.57e+16	0.000	0.750	0.750
a	2.0000	8.67e-16	2.31e+15	0.000	2.000	2.000
=====						
Omnibus:	0.858	Durbin-Watson:	1.912			
Prob(Omnibus):	0.651	Jarque-Bera (JB):	0.763			
Skew:	-0.062	Prob(JB):	0.683			
Kurtosis:	3.054	Cond. No.	24.4			
=====						

Figure 3.7 – The results of the model with a non-linear term

Figure 3.7 presents the results of our regression analysis. The coefficient for X is marked as x , the coefficient for X^2 as x^2 , and the coefficient for A as a . If we compare the coefficient values to the true coefficients in our SCM, we can notice that they are *exactly* the same! This is because we modeled Y as a deterministic function of X and A , not adding any noise.

We can also see that the model correctly decoded the coefficient for the non-linear term (X^2). Although the relationship between X^2 and Y is non-linear, they are related by a linear functional assignment.

Regression and causal effects

We've seen an example of linear regression with a single predictor and an example with an additional quadratic term. In the purely linear case, the *causal* interpretation of the regression result is straightforward: the coefficient of X represents the magnitude of the causal impact of X on Y .

To solidify our intuition, let's see an example.

If the coefficient of X is equal to 6.7, we expect that by increasing the value of X by 1, we will see an increase of 6.7 in Y .

This also generalizes to multiple regression. If we had three predictors, X_1 , X_2 , and X_3 , each of their respective coefficients would have an analogous interpretation. The coefficient of X_1 would quantify the causal effect of X_1 on Y (holding everything else constant) and so on.

When we add a *transformed* version of a variable to the model (as we did by adding X^2), the interpretation becomes slightly less intuitive.

To understand how to interpret the coefficients in such a model, let's get back to the simple univariate case first.

Let's represent the univariate case with the following simplified formula (we omit the intercept and noise):

$$Y = \beta_1 X_1$$

We said before that the coefficient quantifies the magnitude of X_1 's impact on Y . This can be understood as a derivative of Y with respect to X_1 .

If you know a thing or two about calculus, you will quickly notice that this derivative is equal to β_1 – our coefficient. If you don't, don't worry.

Now, let's add the quadratic term to our equation:

$$Y = \beta_1 X_1 + \beta_2 X_1^2$$

Taking the derivative of this expression with respect to X_1 will give us the following:

$$\frac{dY}{dX_1} = \beta_1 + 2\beta_2 X_1$$

Note that we can get this result by applying a technique called the **power rule** (<https://bit.ly/MathPowerRule>) to our equation.

An interesting thing is that now the magnitude of X_1 's impact on Y depends on the value of X_1 . It's no longer quantified solely by its coefficient. When the effect of a variable on the outcome depends on this variable's value, we say that the effect is **heterogeneous**. We'll discuss heterogeneous effects in *Part 2, Causal Inference*.

Note that another perspective on the quadratic term is that this is a special case of interaction between X_1 and itself. This is congruent with the multiplicative definition of interaction that we presented earlier in this chapter. In this light, the quadratic term can be seen as $X_1 \cdot X_1$.

To summarize, adding non-linear terms to linear regression models makes the interpretation of these models more difficult (although not impossible). This is true in both – associational and causal – approaches.

The causal interpretation of linear regression *only holds* when there are *no spurious relationships* in your data. This is the case in two scenarios: when you control for a set of all necessary variables (sometimes this set can be empty) or when your data comes from a properly designed randomized experiment.

Any time you run regression analysis on arbitrary real-world observational data, there's a significant risk that there's hidden confounding in your dataset and so causal conclusions from such analysis are likely to be (causally) biased.

Additionally, remember that in order to obtain a valid regression model, a set of core regression assumptions should be met (linearity in parameters, homoscedasticity of variance, independence of observations, and normality of Y for any fixed value of X). To learn more about these assumptions, check out Westfall & Arias (2022, pp. 17-19).

Wrapping it up

That was a lot of material! Congrats on reaching the end of *Chapter 3*!

In this chapter, we learned about the links between regression, observational data, and causal models. We started with a review of linear regression. After that, we discussed the concept of statistical control and demonstrated how it can lead us astray. We analyzed selected recommendations regarding statistical control and reviewed them from a causal perspective. Finally, we examined the links between linear regression and SCMs.

A solid understanding of the links between observational data, regression, and statistical control will help us move freely in the world of much more complex models, which we'll start introducing in *Part 2, Causal Inference*.

We're now ready to take a more detailed look at the graphical aspect of causal models. See you in the next chapter!

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